

HOW DO I?

How do I manage severe postpartum hemorrhage?

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In the United States, postpartum hemorrhage (PPH) accounts for 4.6% of all maternal deaths and is responsible for major peripartum medical and surgical morbidity. Therefore, a national health priority is to ensure that women who experience severe PPH receive timely, appropriate, and effective treatment. In this article, we describe our system-wide approach for the planning and delivery of women with suspected placenta accreta spectrum disorder, a condition associated with life-threatening blood loss at the time of delivery. We also highlight current evidence related to transfusion decision making and hemostatic monitoring during active postpartum bleeding. Specifically, we describe how we activate and use the massive transfusion protocol to obtain sufficient volumes and types of blood products. We also describe how we use viscoelastic monitoring (thromboelastography) and standard laboratory tests to assess the maternal coagulation profile. Finally, we review the findings of recent studies examining the potential efficacy of tranexamic acid and fibrinogen concentrate as adjuncts for PPH prevention and treatment. We describe how we have incorporated these drugs into PPH treatment protocols at our institution.

Severe postpartum hemorrhage (PPH), characterized by major bleeding after delivery, can result in devastating outcomes for women after delivery. PPH is responsible for 4.6% of all US maternal deaths¹ and is a leading cause of major medical and surgical morbidity, including organ failure from hypoperfusion, disseminated intravascular coagulation, and hysterectomy.²⁻⁵ Consequently, a national maternal health priority is to improve how maternal health care providers prevent and manage severe PPH,⁶ especially as US rates of PPH and PPH-related morbidity have risen in recent years.^{7,8}

In this “How Do I?” article, we describe how we treat women with severe PPH at our institution, Lucile Packard Children’s Hospital, a tertiary care obstetric center in Stanford, California. Given that PPH preparedness can vary when bleeding is anticipated versus unanticipated,⁹ we discuss our approach to preoperative planning, including blood ordering for patients with placenta accreta spectrum (PAS) disorder, a condition that can result in anticipated life-threatening blood loss. We also discuss our approaches to the active management of severe unanticipated PPH, with a focus on transfusion decision making; hemostatic monitoring, including point-of-care technologies; and the use of prohemostatic drugs, specifically fibrinogen concentrate and tranexamic acid. For each topic, we provide a synopsis

ABBREVIATIONS: APTT = activated partial thromboplastin time; FFP = fresh frozen plasma; MTP = massive transfusion protocol; PAS = placenta accreta spectrum; PLTs = platelets; PPH = postpartum hemorrhage; PT = prothrombin time; ROTEM = rotational thromboelastometry; TEG = thromboelastography; TXA = tranexamic acid.

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of key literature followed by a description of our institutional practice.

PATIENTS WITH PLACENTA ACCRETA SPECTRUM DISORDER

In this section, we describe how we plan delivery for women with suspected PAS. Based on US administrative data, the incidence of PAS increased from 1.6 to 2.0 per 1000 repeat cesarean deliveries between 2000 and 2011, a 31% increase over time.¹⁰ This increase is likely related to a concomitant increase in the repeat cesarean delivery rate (from 8.7% to 15.3%) over the same time period.¹⁰ The incidence of PAS varies in other well-developed countries,¹¹ which may be related to variation in both case ascertainment and national rates of cesarean delivery.

PAS is one of the most dangerous conditions associated with pregnancy because of the risk of life-threatening hemorrhage.¹² The increased PPH risk is related to abnormal placental invasion into the myometrium. PAS classification is based on the depth of this invasion. Placenta accreta occurs when the villi are directly attached to the myometrium; placenta increta when the villi extend into the myometrium; and placenta percreta when the villi extend through the myometrium and beyond uterine serosa, often into adjacent viscera and organs, most commonly the bladder. Women may experience extremely high blood loss at or soon after delivery, not infrequently ranging from 2000 to 6000 mL.^{13,14} Consequently, transfusion requirements are high. For example, in a single-center study of 66 cases of women with PAS, 63 (95%) women received red blood cells (RBCs), of whom 26 (39%) received more than 10 units and 7 (11%) received more than 20 units.¹⁵ Population-level data confirm that PAS is the most common etiology for massive transfusion in obstetrics. Among women undergoing delivery hospitalization in New York State between 1998 and 2007, PAS accounted for 26% of all massive transfusion events, classified by the receipt of 10 or more units of blood.¹⁶

How do we plan for these deliveries?

Our hospital is a referral center for women with suspected PAS. The core multidisciplinary team includes maternal-fetal medicine physicians, obstetric anesthesiologists, gynecologic oncologists, and neonatologists. When these women are admitted to the hospital, the team notifies transfusion services physicians and seeks input from medical specialists if any woman has moderate to severe medical comorbidity. Care is coordinated through the Program for Placental Disorders at Stanford Medical Center. Because maternal outcomes are improved if delivery occurs in a hospital with a high volume of cases managed by an experienced multidisciplinary team,¹⁷ experts recommend delivery in a "Center of Excellence" for women with suspected PAS.¹⁸ Blood

services criteria for a PAS Center of Excellence include massive transfusion capabilities, availability of cell saver, perfusionists, experience with alternative blood products, and guidance from transfusion medicine specialists.¹⁸

Based on a Joint Consensus Opinion Statement from the American College of Obstetricians and Gynecologists and the Society for Maternal-Fetal Medicine,¹⁹ a scheduled cesarean delivery is typically planned between 34 and 36 weeks' gestation. In our institution, inpatient admission occurs at approximately 32 weeks' gestation to allow for patient proximity, close patient monitoring, multidisciplinary team review, and betamethasone administration for fetal lung maturity. Anemic patients undergo additional testing for iron deficiency, often receiving intravenous iron supplementation if ferritin levels are low to optimize RBC mass and hemoglobin levels before delivery. Given the gestational age at the time of admission, oral iron is not preferred as there may be insufficient time to expect a hemoglobin rise before delivery. Previous reviews discuss general treatment approaches for iron deficiency and iron deficiency anemia (oral and intravenous iron, erythropoiesis stimulating agents, and RBCs) in pregnant patients.^{20,21}

In the week before surgery, a maternal-fetal medicine physician securely emails physicians and nurses in all relevant specialties (including obstetrics, obstetric anesthesiology, gynecology-oncology, trauma surgery, transfusion medicine, vascular surgery, interventional radiology, neonatology, and pathology) about the date and time of the cesarean delivery and key patient information. We schedule surgery in the Stanford Hospital's main operating room during the assigned gynecologic oncologist's surgical block time. At our institution, gynecologic oncologists are always involved because of their expertise and experience with operating in the retroperitoneal or "deep pelvic" space, especially during periods of rapid blood loss. Our practice is consistent with Center of Excellence recommendations for PAS,¹⁸ with a pelvic surgeon (i.e., a gynecologic oncologist or urogynecologist) as a core member of a multidisciplinary team.

On the day of surgery, the anesthesia team uses a preoperative checklist. The checklist contains a list of essential equipment, drugs, and key clinical reminders (see Appendix for checklist components). Key equipment includes a rapid infusor device (Rapid Infusor RI-2; Belmont Medical Technologies); arterial line and central venous pressure transducer sets; programmable infusion pumps for infusing drugs, for example, vasopressors and oxytocin; fluid-warming devices and forced-air warming machines; a transesophageal echocardiography machine; and a cell saver. Hypocalcemia has been reported during large blood losses requiring 4 or more RBC units in women undergoing cesarean hysterectomy for PAS.²² Therefore, before each scheduled cesarean hysterectomy for suspected PAS, we obtain 10 × 1 g minijets of calcium gluconate and maintain patients' ionized calcium level at >0.9 mmol/L. We also

closely monitor ionized calcium levels from arterial blood gas samples in women with other bleeding etiologies.

What is our approach to preoperative blood ordering?

Based on data from population-wide studies and single-center studies, patients with PAS experience high intraoperative blood losses and require substantial blood product support, especially in women with the most invasive placental subtypes (increta and percreta).^{13–16,23} In preparation for major bleeding, the blood bank performs a preoperative blood type and antibody screen before the day of surgery and alerts the care team if any significant antibodies are identified. For each woman, we order 12 units of type-compatible cross-matched RBCs, 8 units of type-compatible plasma, and 2 apheresis units of ABO-compatible platelets, independent of the suspected PAS subtype. On the day of surgery, the transfusion services department sends blood products to the operating room in modified temperature-controlled coolers (Igloo). At our institution, introduction of these modified coolers led to a 60% and 28% reduction in monthly plasma and RBC wastage across the adult hospital, respectively.²⁴

Examining blood product usage in women with PAS allows transfusion services departments to audit and modify their internal approaches for preparing and delivering blood products to care teams at the time of delivery. We reported blood product use in 136 women who delivered at our institution between 2009 and 2014 with pathologically confirmed PAS.²⁵ Key patient characteristics and data for blood products issued and transfused are presented in Table 1. As expected, transfusion use was high among women with placenta accreta, increta, and percreta (71%, 82%, and 82%, respectively). Although the requirement for additional blood products from our massive transfusion protocol (MTP) was low

among women with the most severely invasive subtypes (2% of women with placenta increta and percreta received O-negative RBCs), 40% of women who received RBCs from the MTP were not suspected of PAS during the antenatal period. In addition, PAS is one of the most common hemorrhage etiologies requiring ongoing blood support via an obstetric MTP.^{26,27} These data highlight the importance of having an MTP at an institution providing delivery services for women with PAS. Ongoing blood product support from an MTP can be potentially lifesaving if crossmatched blood products are rapidly depleted during torrential intraoperative hemorrhage. We discuss the indications and clinical value of an obstetric-specific MTP in the next section.

KEY APPROACHES TO MANAGING ACTIVE BLEEDING IN PATIENTS WITH MAJOR PPH

How do we monitor hematologic and coagulation indices?

During major postpartum blood loss, we often use thromboelastography (TEG; Hemonetics Corp.) as well as standard laboratory tests to assess patients' hematologic and coagulation, electrolyte, and acid-base profiles. Based on expert reviews and guidelines, viscoelastic hemostatic assays (TEG or rotational thromboelastometry [ROTEM]; Instrumentation Laboratory) can detect coagulopathy, observe trends in the maternal coagulation profile over time, and differentiate surgical bleeding from coagulopathic bleeding.^{28–32} These assays reflect cell-based models of hemostasis, providing data of the dynamic aspects of clot formation (including clot initiation, propagation, and amplification) and clot lysis in whole blood. Prior reviews describe TEG and ROTEM mechanics and measured parameters.^{33–35}

TABLE 1. Key patient characteristics and blood product data in a cohort of women with pathologically confirmed placenta accreta syndrome who delivered at Stanford University Medical Center between 2009 and 2014*

	Microscopic placenta accreta (n = 39)	Placenta accreta (n = 21)	Placenta increta (n = 17)	Placenta percreta (n = 17)
Previous cesarean delivery	8 (21)	11 (52)	13 (76)	17 (100)
Placenta previa in the index pregnancy	7 (18)	12 (57)	13 (76)	16 (94)
Hysterectomy performed after delivery	1 (3)	16 (76)	16 (94)	17 (100)
Patients issued blood products	17 (44)	19 (90)	17 (100)	17 (100)
Number of units issued per patient*:				
RBCs	4 [2–12]	12 [6–24]	13 [7–18]	16 [6–27]
Plasma	1 [0–4]	4 [4–12]	10 [4–12]	8 [4–14]
Platelets	0 [0–1]	2 [1–3]	1 [1–2]	2 [1–3]
Cryoprecipitate	0 [0–0]	0 [0–0]	0 [0–1]	1 [0–4]
Patients transfused with any blood product	11 (28)	15 (71)	14 (82)	14 (82)
Number of units transfused among patients issued blood products				
RBCs	2 [0–4]	4 [2–14]	6 [1–10]	8 [2–13]
Plasma	2 [0–4]	4 [0–12]	4 [0–8]	6 [4–11]
Platelets	0 [0–1]	1 [0–3]	0 [0–2]	1 [1–3]
Cryoprecipitate	0 [0–0]	0 [0–1]	0 [0–1]	0 [0–2]

Data presented as n (%) and median [interquartile range].

* Modified table from Brookfield et al.²³

These assays have several potential benefits over standard plasma coagulation tests. First, it is uncertain whether standard coagulation tests (prothrombin time [PT] and activated partial thromboplastin time [APTT]) accurately reflect hemostatic changes during active bleeding. PT and APTT are poor predictors of bleeding, correlate weakly with clinically relevant bleeding, and typically remain in the normal range in women who experience severe PPH as well as the majority of women who require massive transfusion (≥ 8 units of RBCs within 24 hr of delivery).^{32,36–39} Second, compared with TEG or ROTEM, turnaround times for PT, APTT, and Clauss fibrinogen levels are unacceptably long.^{40,41} At our institution, a “super-stat” coagulation screen turnaround time is 45 minutes. These delays can often result in the care team making transfusion decisions based solely on clinical judgment. Clauss fibrinogen levels may be falsely elevated if starch solutions have been infused during fluid resuscitation.⁴² We accept that some institutions may have developed a rapid emergency hemorrhage panel as an alternative to point-of-care testing for evaluating hemostasis in bleeding patients. At one institution, immediate testing and shortened centrifuge times resulted in mean turnaround times of 14 minutes (range, 6–28 minutes).⁴³

TEG and ROTEM assays provide graphic information of the kinetics of clot formation and lysis in whole blood of women who experience PPH. In a study assessing TEG changes in 49 women with major obstetric hemorrhage, clot stability was decreased (based on alpha angle and maximum amplitude values), fibrinolysis was decreased (based on LY30 values), and clot initiation was increased (based on reaction times) compared to 49 women without hemorrhage.⁴⁴ Studies of women with PPH in which ROTEM was used also reveal important changes in the maternal coagulation profile. In a study of 37 women with severe PPH, ROTEM parameters of the FIBTEM test were altered, namely, prolonged clotting times, reduced clot amplitude at 5 minutes (A5) and 15 minutes, and reduced maximum clot firmness (MCF); these parameters correlated well with plasma fibrinogen levels.⁴⁵ In a separate study of 356 women with PPH, a FIBTEM A5 less than 10 mm was associated with more prolonged bleeding, blood transfusion, and the duration of stay in the high-dependency unit.⁴⁶ Although large trials comparing transfusion strategies guided by TEG or ROTEM versus standard laboratory indices are lacking, findings from a meta-analysis of 17 trials, mostly involving cardiac surgery, suggested that TEG or ROTEM may reduce the need for blood products. In this analysis, TEG or ROTEM significantly reduced the proportion of patients who received RBCs (relative risk [RR], 0.86; 95% confidence interval [CI], 0.79–0.94), fresh frozen plasma (FFP; RR, 0.57; 95% CI, 0.33–0.96), and platelets (PLTs; RR, 0.73; 95% CI, 0.60–0.88).

Other important advantages of TEG and ROTEM are the short turnaround times. At our hospital, we estimate turnaround times of 20 minutes to obtain results from TEG after sending samples to the hematology laboratory. Housing the

TEG devices in the hematology laboratory allows the care team to focus on providing clinical care, thereby reducing task saturation. The laboratory is also well versed in performing quality control and device maintenance, whereas clinicians may not have the time, interest, or expertise to meet quality standards set by hospital point-of-care committees. In our hematology laboratory, technicians run TEG assays from citrated recalcified blood samples. To limit coagulation activation,⁴⁷ blood samples in sodium citrate blood tubes are sent to the hematology laboratory in padded transport pods via the pneumatic tube transport system. The primary care team visualizes real-time TEG traces via a secure remote viewing system embedded into the electronic medical record (EPIC).

Because few ROTEM and TEG algorithms for managing PPH have been published,^{48,49} we developed an institution-specific TEG algorithm for PPH management (Fig. 1). As yet, we have not performed impact or validation studies to assess our algorithm's effectiveness. However, there are accumulating data suggesting that a ROTEM-guided treatment algorithm for severe PPH may reduce hemorrhage-related morbidity. In an impact study at a tertiary obstetric center in the United Kingdom, compared with a standard shock pack, a massive hemorrhage protocol (comprising a ROTEM-guided algorithm and fibrinogen concentrate) was associated with a significant reduction in blood product use and transfusion-associated circulatory overload and nonsignificant reductions in the rate of intensive care admission and hysterectomy.⁵⁰ Similar findings were reported in a US hospital that transitioned from an empiric hemorrhage protocol to a ROTEM-guided protocol.⁵¹

Do we have a massive transfusion protocol for obstetrics?

When the rate and magnitude of blood loss outpace the time required to prepare and transport crossmatched blood products, activation of an institution-specific MTP ensures timely delivery of sufficient types and volumes of blood products to the primary care team.^{52,53} Our MTP includes 6 units of uncrossmatched, blood group O or type-specific RBCs, 4 units of plasma (thawed or liquid), and 1 apheresis platelet unit.^{52,54} The full complement of blood products is transported to the labor and delivery unit or operating room in coolers within 5 to 10 minutes from a satellite blood bank located in the children's hospital. After establishing surgical and hemostatic control of bleeding, the care team can deactivate the MTP and transition to crossmatched compatible RBCs as needed.

Currently, our approach for MTP activation relies on clinical judgment. This involves joint clinical assessment by the attending obstetrician and anesthesiologist of the rate and magnitude of blood loss, etiology of bleeding, the response to medical and surgical interventions to control bleeding, clinical features of hemorrhage and hypoperfusion

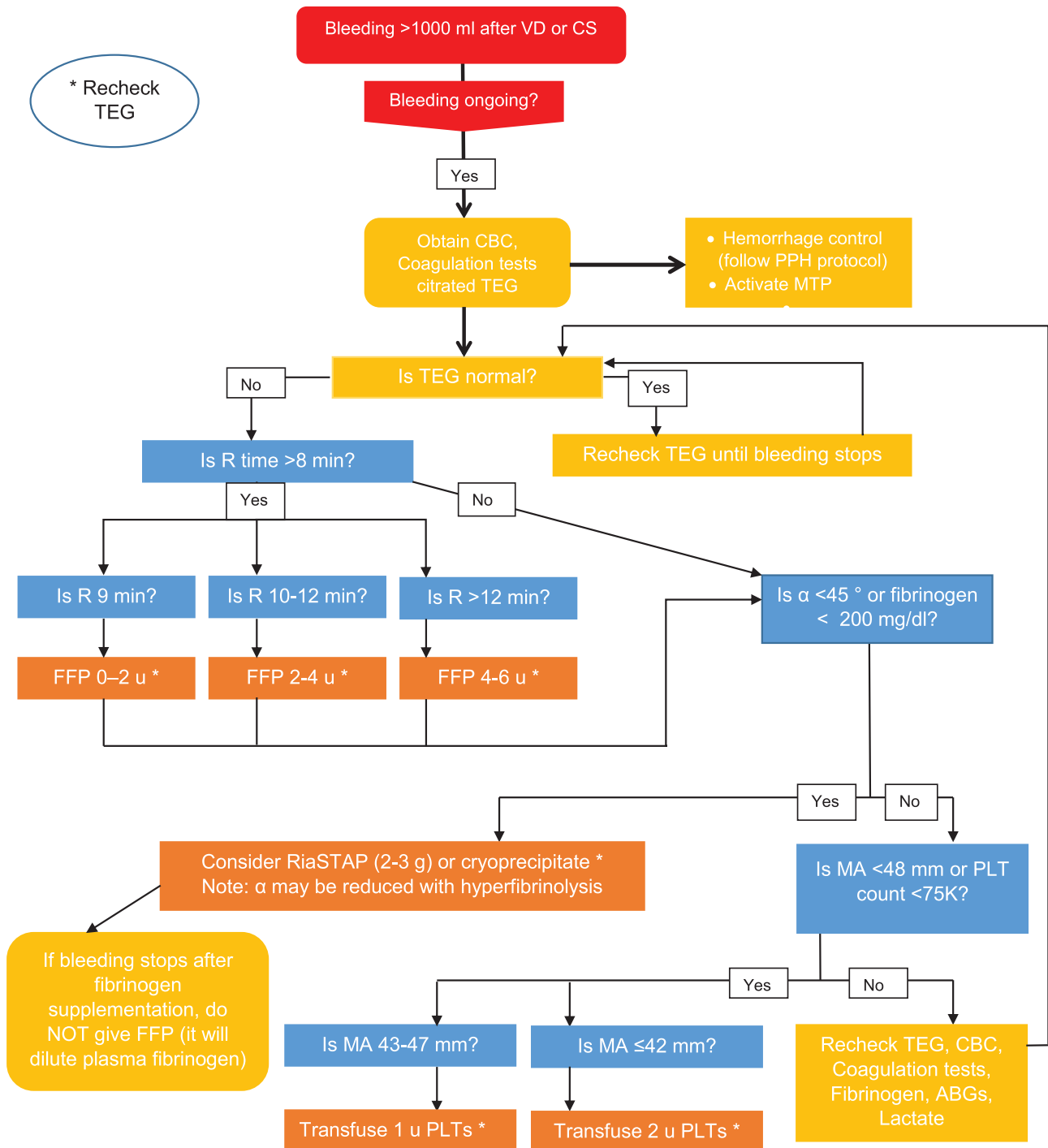


Fig. 1. ABGs = arterial blood gases; CBC = complete blood count; FFP = fresh frozen plasma; MA = maximum amplitude; MTP = massive transfusion protocol; PLTs = platelets; PPH = postpartum hemorrhage; R time = reaction time; RiASTAP = fibrinogen concentrate; TEG = thromboelastography. [Color figure can be viewed at wileyonlinelibrary.com]

(including tachycardia, tachypnea, hypotension, altered mental status, and reduced urine output), degree of cardiovascular support, and likelihood of continual blood loss.

As yet, no studies have examined clinical prediction tools for MTP activation in obstetrics.⁵⁵ Given the lack of

validated criteria for defining massive obstetric hemorrhage and the concerns about the accuracy, timing, and clinical relevance of blood loss estimation, care teams may consider the “intensity of resuscitation,” a proxy of hemorrhage severity, to initiate a MTP. The idea originates from the trauma

literature that links receipt of 4 or more units of any fluid within 30 minutes of hospital admission to an increased risk of mortality at 6 and 24 hours. An alternative proposed approach for MTP initiation is the initial transfusion use, with a threshold of 3 to 4 units of RBCs per hour.^{56,57} Further research is needed to examine whether resuscitation intensity is an appropriate measure for determining if and when to activate an MTP for severe PPH.

What is our transfusion approach for PPH management?

Based on evidence from the trauma literature, PPH guidelines published by the American College of Obstetricians and Gynecologists and the California Maternal Quality Care Collaborative recommend using formulaic or fixed ratios of RBCs:FFP:PLTs.^{58,59} However, there are no high-quality trial data in obstetrics to substantiate this recommendation. Furthermore, marked differences exist between the obstetric and trauma populations in baseline physiology and pathophysiology of coagulopathy related to hemorrhage.⁴⁸ Nonetheless, the impact of national guidelines on clinicians' practice, slow turnaround times of laboratory coagulation tests, as well as clinical assessment, may explain high rates of FFP use during severe PPH. For example, in a retrospective study of 1495 French patients with severe PPH, 69% of transfused patients received FFP.⁶⁰

It is unclear whether early FFP transfusion leads to improved maternal outcomes. In an observational study of 1216 women with persistent PPH (defined as ≥ 1000 mL blood loss refractory to first-line intervention), FFP transfused within 60 minutes of persistent PPH onset was not associated with reduced severe hemorrhage-related morbidity (defined as a composite of death, hysterectomy, or arterial embolization) compared with no or later FFP transfusion.⁶¹ In a different observational study, the restrictive use of FFP in 605 women with a moderate to severe PPH and a FIBTEM A5 greater than 15 mm did not result in ongoing bleeding secondary to hemostatic failure.⁶² Furthermore, the majority of women with severe PPH typically have normal coagulation profiles and platelet counts when blood products are administered.^{37,38,46,63} Given that the fibrinogen concentration of FFP (median concentration = 2 mg/mL; range 0.8–3 mg/mL)⁶⁴ may be lower than patients' plasma fibrinogen concentration, FFP may paradoxically lower plasma fibrinogen levels in women who have moderate PPH and normal coagulation indices.⁶⁵ Other potential complications associated with high-volume FFP use include hypervolemia, transfusion-associated circulatory overload, and transfusion-related acute lung injury (the most common cause of transfusion-related death in the United States).⁶⁶

Any decision to transfuse PLTs warrants careful consideration. A retrospective study of 347 women with moderate to severe PPH reported a low overall incidence of PLT transfusion (3.4%), with a PLT count less than $75 \times 10^9/L$

recorded in only 2% women.⁶⁷ PLT transfusion was more likely in women with thrombocytopenia before PPH onset or consumptive coagulopathies from amniotic fluid embolism or placental abruption. Women with an overall blood loss less than 5000 mL due to uterine atony or trauma were not associated with having a postpartum PLT count less than $75 \times 10^9/L$. Therefore, the use of a fixed-ratio approach would have exposed many women with PLT counts above a threshold level of $75 \times 10^9/L$ to unnecessary PLT transfusions. However, in a study examining hematologic indices of 108 women who received massive transfusion, the median lowest PLT count was $68 \times 10^9/L$ (interquartile range = 50–95 $\times 10^9/L$).³⁹ Therefore, we advise our physicians to maintain a PLT count greater than $50 \times 10^9/L$ during severe PPH.

Despite the uncertainty of maternal benefit associated with formulaic ratios of RBC:FFP:PLTs for severe PPH, at our institution, the primary care team, primarily obstetric anesthesiologists, are the final arbiters of transfusion decision making. Independent of the chosen transfusion strategy, the care team sends superstat laboratory tests and TEG assays early and at regular intervals (every 20–30 min) during active bleeding. Although “real-time” results are not possible, laboratory and TEG indices still provide valuable trend data to assess the effectiveness of transfusion decision making, confirm adequate hemostasis after the surgical arrest of bleeding and may help identify or rule out coagulopathy.

The presence and extent of coagulation disturbance can vary according to PPH etiology. Compared with other PPH etiologies, observational data suggest that women with placental abruption can experience the largest fall in PLT count during massive transfusion,³⁹ the highest rate of PLT transfusion during severe PPH,⁶⁷ and significantly lower ROTEM FIBTEM A5 and prolonged EXTEM clotting times.⁵⁰ On the basis of this evidence, we closely monitor all women with suspected placental abruption for possible consumptive coagulopathy before and after delivery.

Do we use fibrinogen concentrate?

Fibrinogen concentrate has attracted considerable interest as a therapy for treating hypofibrinogenemia in bleeding patients. Prior reviews provide detailed descriptions of fibrinogen concentrate, including its formulation, licensing, pharmacokinetics, and studies of effectiveness in the nonobstetric setting.^{42,66,68} In summary, fibrinogen concentrate is derived from human plasma, undergoes viral inactivation, is available in the form of pasteurized, lyophilized product and does not require crossmatching. It is reconstituted in sterile water to a concentration of 20 mg/mL. Each vial contains approximately 1 g of fibrinogen, with a range of 900–1300 mg (according to the package insert). It is normally injected intravenously (at a rate of 5 mL/min); however, it is possible to infuse more quickly when severe bleeding occurs.⁶⁶

Given that hypofibrinogenemia is the primary indication, determining if and when to use fibrinogen concentrate

during PPH is important. During the early phase of postpartum bleeding, a Clauss fibrinogen level less than 200 mg/dL has been shown to be an important risk factor for progression to severe hemorrhage and increased blood product requirement.^{46,63,69,70} Based on expert opinion and guidelines from the Scientific and Standardization Committee of the International Society for Thrombosis and Hemostasis, the Royal College of Obstetricians and Gynecologists (United Kingdom), and the Association of Anesthetists of Great Britain and Ireland, a target fibrinogen level of at least 200 mg/dL has been recommended during active obstetric bleeding, even if the PT and APTT indices are normal.^{31,42,48,66,71,72} In contrast, US anesthesia and obstetric societies' guidelines do not specifically state a target fibrinogen level for PPH management.^{58,73}

A number of studies have evaluated the impact of fibrinogen supplementation with fibrinogen concentrate in women with severe PPH. In a randomized trial of 249 women with early PPH, Wikkelsø et al.⁷⁴ compared 2 g of fibrinogen concentrate versus saline as preemptive treatment. The frequency of RBC transfusion use did not differ statistically between women receiving fibrinogen concentrate versus saline (20% vs. 22%; $p = 0.88$). The lack of effect may be because few women (2.2%) had a baseline

fibrinogen concentration less than 200 mg/dL.⁷⁴ Using ROTEM technology, Collins et al.⁷⁵ performed a multicenter study to compare the number of RBC units transfused among 55 women with PPH and a FIBTEM A5 of 15 mm or less randomized to fibrinogen concentrate versus placebo. Rates of transfusion did not significantly differ between groups (median [interquartile range] RBC units = 1 [0-2] vs. 1 [0-2]; fibrinogen concentrate vs. placebo, respectively). A prespecified subgroup analysis suggested that a FIBTEM A5 greater than 12 mm or a fibrinogen level greater than 200 mg/dL may be adequate for hemostasis and not require fibrinogen supplementation. However, several observational studies from single obstetric centers suggest fibrinogen concentrate use is associated with higher plasma fibrinogen levels, lower blood loss values, and reduced FFP usage.^{76,77}

At our institution, we currently stock fibrinogen concentrate vials in temperature-controlled refrigerators on the labor and delivery unit. This allows members of the care team to access and use fibrinogen concentrate immediately. Therefore, we consider fibrinogen concentrate as an adjunctive therapy for PPH in an actively bleeding patient with a fibrinogen level less than 200 mg/dL and/or a TEG alpha angle less than 45° (see Figs. 1 and 2). When teams are called to assess and manage unanticipated PPH, fibrinogen

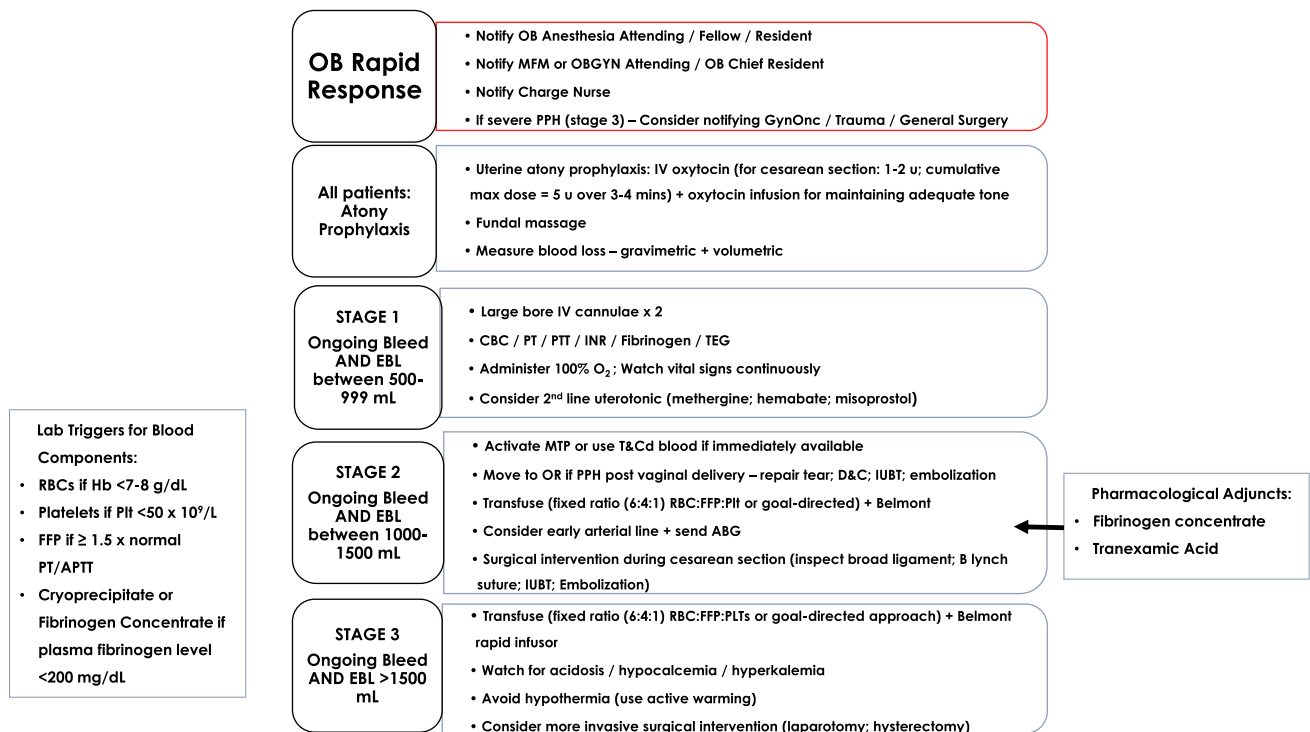


Fig. 2. ABG = arterial blood gas; CBC = complete blood count; CS = cesarean section; FFP = fresh frozen plasma; GynOnc = gynecological oncologist; Hb = hemoglobin; INR = international normalized ratio; IV = intravenous; IUBT = intrauterine balloon tamponade; MFM = maternal-fetal medicine; OB = obstetric; OBGYN = obstetrics and gynecology; OR = operating room; PLT = platelets; PPH = postpartum hemorrhage; PT = prothrombin time; PTT = partial thromboplastin time; TEG = thromboelastography; T&C = type and cross; VD = vaginal delivery. [Color figure can be viewed at wileyonlinelibrary.com]

concentrate may be considered empirically when the cumulative blood loss is estimated to be at least 1000 to 1500 mL in a patient with active bleeding (Fig. 2). However, if time permits, it is preferable to identify significant hypofibrinogenemia or reduced fibrinogen contribution to clot formation (with TEG or ROTEM) before administering fibrinogen concentrate.

Do we use tranexamic acid?

Recent evidence suggests that tranexamic acid (TXA), an antifibrinolytic agent that competitively inhibits plasminogen activation, is an important therapeutic adjunct in women with PPH. The World Maternal Antifibrinolytic (WOMAN) trial, published in the *Lancet* in 2017, was a pragmatic, international, double-blind, placebo-controlled, multicenter study that randomized 20,060 women with PPH to receive 1 to 2 g of intravenous TXA versus placebo in addition to usual care.⁷⁸ This study involved nearly 200 hospitals in 21 countries. The main findings indicated a small reduction in the risk of death from exsanguination in women assigned TXA versus placebo (1.5% vs. 1.9%; RR, 0.81, 95% CI, 0.65-1.0; $p = 0.045$). Treatment benefit appeared to be strongest if TXA is administered within 3 hours of birth (deaths from exsanguination were 1.2% vs. 0.7% in the TXA vs. placebo groups, respectively). Authors from the WOMAN trial concluded that TXA reduced the risk of death from exsanguination without adverse effects. Based on the WOMAN trial findings, the World Health Organization recommends that all women with PPH after vaginal or cesarean delivery receive TXA within 3 hours of birth in addition to standard care.⁷⁹

Several aspects of the WOMAN trial require close attention. First, the majority of study sites were in low-income countries; therefore, generalizability of the findings to hospitals to high-income countries is uncertain. Moreover, the death rate from exsanguination among women with PPH is substantially lower in the United States (approx. 38 per 100,000 PPH events)⁸⁰ than reported in the placebo group in the WOMAN trial, a nearly 40-fold difference. Second, causes of death were documented by participating clinicians and not from autopsy reports. Third, the effect of TXA was of borderline statistical significance,⁸¹ with a number needed to treat of 250. Fourth, TXA was not associated with a reduction in rates of surgical or radiological interventions (including hysterectomy, intrauterine tamponade, embolization, or arterial ligation). Further, rates of transfusion (54% in both groups) and the mean number of blood units transfused did not significantly differ between groups. This raises further questions about the potential impact of TXA in reducing PPH-related morbidity among women who deliver in well-resourced and well-staffed hospitals in the developed world.^{82,83}

At our institution, we have included TXA into our current PPH protocol as an adjunctive, and not compulsory, drug therapy (Fig. 2). The decision to use TXA is at the

discretion of the care team on the labor and delivery unit. The maximum dose of TXA is 2 g because of concern that higher doses may increase the risk of renal cortical necrosis in hypoperfused patients.⁸³

Limited high-quality data exist to determine the risk-benefit of administering TXA as PPH prophylaxis before vaginal or cesarean delivery. The largest and most robust randomized trial (the Tranexamic Acid for Preventing Postpartum Hemorrhage Following a Vaginal Delivery [TRAAP] trial) examined whether prophylactic TXA, in addition to prophylactic oxytocin, reduces the risk of PPH after vaginal delivery compared with placebo in 3891 women. The main finding was that risk of PPH did not differ between groups (TXA, 8.1% vs. placebo, 9.8%; RR, 0.83; 95% CI, 0.68-1.01; $p = 0.07$).⁸⁴ However, in a subgroup analysis, TXA significantly reduced the PPH risk among women with operative vaginal delivery (9.4% vs. 14.7%; $p = 0.04$). The incidence of thrombotic events did not differ between groups (TXA, 0.1% vs. placebo, 0.2%; $p = 0.37$). Nausea and vomiting were more common among women with TXA (7% vs. 3.2%; $p < 0.001$). Other trials assessing TXA for PPH prophylaxis during vaginal or cesarean delivery are predominantly small and low quality with major methodological issues.⁸⁵ Recent meta-analyses differ in their assessments of trial design characteristics, for example, allocation concealment,^{86,87} suggesting that the risk of bias of randomized trials assessing TXA efficacy in these meta-analyses is not being consistently reported.

Therefore, based on the negative findings of the TRAAP trial and the uncertain quality of other published trials investigating TXA as prophylaxis, we do not currently use TXA as prophylaxis against PPH for women undergoing vaginal or cesarean delivery, even for women who are at high risk for PPH. We acknowledge that centers with a limited blood bank inventory and no fibrinogen concentrate in their drug formulary may have a lower threshold for using TXA for severe PPH treatment.

CONCLUSION

In this review, we describe our institutional approaches to the preoperative planning for women with suspected PAS awaiting scheduled cesarean hysterectomy. We highlight the importance of timely blood product support and the potential value of viscoelastic monitors when managing women with severe and active postpartum bleeding. Finally, we provide an overview of how we use fibrinogen concentrate and TXA in our management of severe PPH.

CONFLICT OF INTEREST

AJB and DJL have disclosed no conflicts of interest. LTG is a consultant for American Regent, Inc, which manufactures Venofer (iron sucrose).

REFERENCES

1. Hoyert DL, Minino M. Maternal mortality in the United States: Changes in coding, publication, and data release, 2018. *Nat Vital Stat Rep*. Hyattsville, MD: National Center for Health Statistics; 2020.
2. Bateman BT, Mhyre JM, Callaghan WM, et al. Peripartum hysterectomy in the United States: nationwide 14 year experience. *Am J Obstet Gynecol* 2012;206:63 e1-8.
3. Bateman BT, Berman MF, Riley LE, et al. The epidemiology of postpartum hemorrhage in a large, nationwide sample of deliveries. *Anesth Analg* 2010;110:1368-73.
4. Wanderer JP, Leffert LR, Mhyre JM, et al. Epidemiology of obstetric-related ICU admissions in Maryland: 1999-2008*. *Crit Care Med* 2013;41:1844-52.
5. Al-Zirqi I, Vangen S, Forsen L, et al. Prevalence and risk factors of severe obstetric haemorrhage. *BJOG* 2008;115:1265-72.
6. Main EK, Goffman D, Scavone BM, et al. National partnership for maternal safety: consensus bundle on obstetric hemorrhage. *Obstet Gynecol* 2015;126:155-62.
7. Reale SC, Easter SR, Xu X, et al. Trends in postpartum hemorrhage in the United States from 2010 to 2014. *Anesth Analg* 2020;130:e119-22.
8. Fingar KR, Hambrick MM, Heslin KC, et al. Trends and disparities in delivery hospitalizations involving severe maternal morbidity, 2006-2015 healthcare cost and utilization project (HCUP) statistical brief #243. Rockville, MD: Agency for Healthcare Research and Quality (US); 2018.
9. Gallos G, Redai I, Smiley RM. The role of the anesthesiologist in management of obstetric hemorrhage. *Semin Perinatol* 2009;33:116-23.
10. Creanga AA, Bateman BT, Butwick AJ, et al. Morbidity associated with cesarean delivery in the United States: is placenta accreta an increasingly important contributor? *Am J Obstet Gynecol* 2015;213:384 e1-11.
11. Jauniaux E, Chantaine F, Silver RM, et al. FIGO consensus guidelines on placenta accreta spectrum disorders: epidemiology. *Int J Gynaecol Obstet* 2018;140:265-73.
12. Silver RM, Branch DW. Placenta accreta spectrum. *N Engl J Med* 2018;378:1529-36.
13. Shaanker S, Shamshirsaz A, Haviland M, et al. Utilization and outcomes of massive transfusion protocols in women with and without invasive placentation. *J Matern Fetal Neonatal Med* 2019;1:1-5.
14. Wright JD, Pri-Paz S, Herzog TJ, et al. Predictors of massive blood loss in women with placenta accreta. *Am J Obstet Gynecol* 2011;205:38 e1-6.
15. Stotler B, Padmanabhan A, Devine P, et al. Transfusion requirements in obstetric patients with placenta accreta. *Transfusion* 2011;51:2627-33.
16. Mhyre JM, Shilkut A, Kuklina EV, et al. Massive blood transfusion during hospitalization for delivery in New York State, 1998-2007. *Obstet Gynecol* 2013;122:1288-94.
17. Shamshirsaz AA, Fox KA, Salmanian B, et al. Maternal morbidity in patients with morbidly adherent placenta treated with and without a standardized multidisciplinary approach. *Am J Obstet Gynecol* 2015;212:218 e1-9.
18. Silver RM, Fox KA, Barton JR, et al. Center of excellence for placenta accreta. *Am J Obstet Gynecol* 2015;212:561-8.
19. Society of Gynecologic Oncology, American College of Obstetricians and Gynecologists, The Society for Maternal-Fetal Medicine, Cahill AG, Beigi R, et al. Placenta accreta spectrum. *Am J Obstet Gynecol* 2018;219:B2-B16.
20. Friedman AJ, Shander A, Martin SR, et al. Iron deficiency anemia in women: a practical guide to detection, diagnosis, and treatment. *Obstet Gynecol Surv* 2015;70:342-53.
21. Munoz M, Pena-Rosas JP, Robinson S, et al. Patient blood management in obstetrics: management of anaemia and haematinic deficiencies in pregnancy and in the post-partum period: NATA consensus statement. *Transfus Med* 2018;28:22-39.
22. Erfani H, Fox KA, Belfort MA, et al. Preventing severe hypocalcemia during surgery for placenta accreta spectrum. *Acta Obstet Gynecol Scand* 2019;98:1358-9.
23. Brookfield KF, Goodnough LT, Lyell DJ, et al. Perioperative and transfusion outcomes in women undergoing cesarean hysterectomy for abnormal placentation. *Transfusion* 2014;54:1530-6.
24. Metcalf RA, Baker SA, Goodnough LT, et al. Transportation cooler interventions reduce plasma and RBC product wastage. *Am J Clin Pathol* 2016;146:18-24.
25. Panigrahi AK, Yeaton-Massey A, Bakhtary S, et al. A standardized approach for transfusion medicine support in patients with morbidly adherent placenta. *Anesth Analg* 2017;125:603-8.
26. Gutierrez MC, Goodnough LT, Druzin M, et al. Postpartum hemorrhage treated with a massive transfusion protocol at a tertiary obstetric center: a retrospective study. *Int J Obstet Anesth* 2012;21:230-5.
27. Margarido C, Ferns J, Chin V, et al. Massive hemorrhage protocol activation in obstetrics: a 5-year quality performance review. *Int J Obstet Anesth* 2019;38:37-45.
28. Reale SC, Farber MK. Point-of-care coagulation testing for obstetric hemorrhage: time for a theranostic approach? *Int J Obstet Anesth* 2019;38:1-3.
29. Kozek-Langenecker SA, Ahmed AB, Afshari A, et al. Management of severe perioperative bleeding: guidelines from the European Society of Anaesthesiology: first update 2016. *Eur J Anaesthesiol* 2017;34:332-95.
30. American Society of Anesthesiologists. Practice guidelines for perioperative blood management: an updated report by the American Society of Anesthesiologists task force on perioperative blood management*. *Anesthesiology* 2015;122:241-75.
31. Collins P, Abdul-Kadir R, Thachil J. The Subcommittees on Women's Health Issues in Thrombosis, Haemostasis, on Disseminated Intravascular Coagulation. Management of coagulopathy associated with postpartum hemorrhage: guidance from the SSC of the ISTH. *J Thromb Haemost* 2016;14:205-10.
32. Solomon C, Collis RE, Collins PW. Haemostatic monitoring during postpartum haemorrhage and implications for management. *Br J Anaesth* 2012;109:851-63.

33. de Lange NM, Lance MD, de Groot R, et al. Obstetric hemorrhage and coagulation: an update. *Thromboelastography, thromboelastometry, and conventional coagulation tests in the diagnosis and prediction of postpartum hemorrhage*. *Obstet Gynecol Surv* 2012;67:426-35.
34. Ganter MT, Hofer CK. Coagulation monitoring: current techniques and clinical use of viscoelastic point-of-care coagulation devices. *Anesth Analg* 2008;106:1366-75.
35. Srinivasa V, Gilbertson LI, Bhavani-Shankar K. Thromboelastography: where is it and where is it heading? *Int Anesthesiol Clin* 2001;39:35-49.
36. Haas T, Fries D, Tanaka KA, et al. Usefulness of standard plasma coagulation tests in the management of perioperative coagulopathic bleeding: is there any evidence? *Br J Anaesth* 2015;114:217-24.
37. de Lloyd L, Bovington R, Kaye A, et al. Standard haemostatic tests following major obstetric haemorrhage. *Int J Obstet Anesth* 2011;20:135-41.
38. Gillissen A, van den Akker T, Caram-Deelder C, et al. Coagulation parameters during the course of severe postpartum hemorrhage: a nationwide retrospective cohort study. *Blood Adv* 2018;2:2433-42.
39. Green L, Knight M, Seeney F, et al. The haematological features and transfusion management of women who required massive transfusion for major obstetric haemorrhage in the UK: a population based study. *Br J Haematol* 2016;172:616-24.
40. Haas T, Spielmann N, Mauch J, et al. Comparison of thromboelastometry (ROTEM(R)) with standard plasmatic coagulation testing in paediatric surgery. *Br J Anaesth* 2012;108:36-41.
41. Jeger V, Zimmermann H, Exadaktylos AK. Can RapidTEG accelerate the search for coagulopathies in the patient with multiple injuries? *J Trauma* 2009;66:1253-7.
42. Levy JH, Goodnough LT. How I use fibrinogen replacement therapy in acquired bleeding. *Blood* 2015;125:1387-93.
43. Chandler WL, Ferrell C, Trimble S, et al. Development of a rapid emergency hemorrhage panel. *Transfusion* 2010;50:2547-52.
44. Karlsson O, Jeppsson A, Hellgren M. Major obstetric haemorrhage: monitoring with thromboelastography, laboratory analyses or both? *Int J Obstet Anesth* 2014;23:10-7.
45. Huissoud C, Carrabin N, Audibert F, et al. Bedside assessment of fibrinogen level in postpartum haemorrhage by thrombelastometry. *BJOG* 2009;116:1097-102.
46. Collins PW, Lilley G, Bruynseels D, et al. Fibrin-based clot formation as an early and rapid biomarker for progression of postpartum hemorrhage: a prospective study. *Blood* 2014;124:1727-36.
47. Espinosa A, Ruckert A, Navarro J, et al. Are TEG((R)) results in healthy blood donors affected by the transport of blood samples in a pneumatic tube system? *Int J Lab Hematol* 2016;38:e73-6.
48. Collis RE, Collins PW. Haemostatic management of obstetric haemorrhage. *Anaesthesia* 2015;70(Suppl 1):78-86, e27-8.
49. McNamara H, Mallaiah S. Managing coagulopathy following PPH. *Best Pract Res Clin Obstet Gynaecol* 2019;61:106-20.
50. McNamara H, Kenyon C, Smith R, et al. Four years' experience of a ROTEM((R))-guided algorithm for treatment of coagulopathy in obstetric haemorrhage. *Anaesthesia* 2019;74:984-91.
51. Snegovskikh D, Souza D, Walton Z, et al. Point-of-care viscoelastic testing improves the outcome of pregnancies complicated by severe postpartum hemorrhage. *J Clin Anesth* 2018;44:50-6.
52. Goodnough LT, Daniels K, Wong AE, et al. How we treat: transfusion medicine support of obstetric services. *Transfusion* 2011;51:2540-8.
53. Burtelow M, Riley E, Druzin M, et al. How we treat: management of life-threatening primary postpartum hemorrhage with a standardized massive transfusion protocol. *Transfusion* 2007;47:1564-72.
54. Goodnough LT, Spain DA, Maggio P. Logistics of transfusion support for patients with massive hemorrhage. *Curr Opin Anaesthesiol* 2013;26:208-14.
55. Foster JC, Sappenfield JW, Smith RS, et al. Initiation and termination of massive transfusion protocols: Current strategies and future prospects. *Anesth Analg* 2017;125:2045-55.
56. Moren AM, Hamptom D, Diggs B, et al. Recursive partitioning identifies greater than 4 U of packed red blood cells per hour as an improved massive transfusion definition. *J Trauma Acute Care Surg* 2015;79:920-4.
57. Savage SA, Zarzaur BL, Croce MA, et al. Redefining massive transfusion when every second counts. *J Trauma Acute Care Surg* 2013;74:396-400; discussion -2.
58. American College of Obstetricians and Gynecologists. ACOG. Practice bulletin no. 183: postpartum hemorrhage. *Obstet Gynecol* 2017;130:e168-86.
59. California Maternal Quality Care Collaborative. Obstetric Hemorrhage Toolkit V 2.0. 2015. Available from: https://www.cmqcc.org/ob_hemorrhage/ob_hemorrhage_compendium_of_best_practices
60. Deleu F, Deneux-Tharaux C, Chiesa-Dubruille C, et al. A population-based analysis of French transfusion practices for women experiencing severe postpartum hemorrhage. *Int J Obstet Anesth* 2019 [epub ahead of print].
61. Henriquez D, Caram-Deelder C, le Cessie S, et al. Association of timing of plasma transfusion with adverse maternal outcomes in women with persistent postpartum hemorrhage. *JAMA Netw Open* 2019;2:e1915628.
62. Collins PW, Cannings-John R, Bruynseels D, et al. Viscoelastometry guided fresh frozen plasma infusion for postpartum haemorrhage: OBS2, an observational study. *Br J Anaesth* 2017;119:422-34.
63. Charbit B, Mandelbrot L, Samain E, et al. The decrease of fibrinogen is an early predictor of the severity of postpartum hemorrhage. *J Thromb Haemost* 2007;5:266-73.
64. Levy JH, Szlam F, Tanaka KA, et al. Fibrinogen and hemostasis: a primary hemostatic target for the management of acquired bleeding. *Anesth Analg* 2012;114:261-74.
65. Collins PW, Solomon C, Sutor K, et al. Theoretical modelling of fibrinogen supplementation with therapeutic plasma, cryoprecipitate, or fibrinogen concentrate. *Br J Anaesth* 2014;113:585-95.
66. Levy JH, Welsby I, Goodnough LT. Fibrinogen as a therapeutic target for bleeding: a review of critical levels and replacement therapy. *Transfusion* 2014;54:1389-405.
67. Jones RM, de Lloyd L, Kealaher EJ, et al. Platelet count and transfusion requirements during moderate or severe postpartum haemorrhage. *Anaesthesia* 2016;71:648-56.

68. Fenger-Eriksen C, Ingerslev J, Sorensen B. Fibrinogen concentrate—a potential universal hemostatic agent. *Expert Opin Biol Ther* 2009;9:1325-33.
69. Cortet M, Deneux-Tharaux C, Dupont C, et al. Association between fibrinogen level and severity of postpartum haemorrhage: secondary analysis of a prospective trial. *Br J Anaesth* 2012;108:984-9.
70. Gayat E, Resche-Rigon M, Morel O, et al. Predictive factors of advanced interventional procedures in a multicentre severe postpartum haemorrhage study. *Intensive Care Med* 2011;37:1816-25.
71. Royal College of Obstetricians and Gynecologists. Prevention and management of postpartum haemorrhage: green-top guideline No. 52. *BJOG* 2017;124:e106-49.
72. Klein AA, Arnold P, Bingham RM, et al. AAGBI guidelines: the use of blood components and their alternatives 2016. *Anaesthesia* 2016;71:829-42.
73. Shaylor R, Weiniger CF, Austin N, et al. National and International Guidelines for patient blood management in obstetrics: a qualitative review. *Anesth Analg* 2016;124:216-32.
74. Wikkelsso AJ, Edwards HM, Afshari A, et al. Pre-emptive treatment with fibrinogen concentrate for postpartum haemorrhage: randomized controlled trial. *Br J Anaesth* 2015;114:623-33.
75. Collins PW, Cannings-John R, Bruynseels D, et al. Viscoelastometric-guided early fibrinogen concentrate replacement during postpartum haemorrhage: OBS2, a double-blind randomized controlled trial. *Br J Anaesth* 2017;119:411-21.
76. Matsunaga S, Takai Y, Nakamura E, et al. The clinical efficacy of fibrinogen concentrate in massive obstetric haemorrhage with hypofibrinogenaemia. *Sci Rep* 2017;7:46749.
77. Seto S, Itakura A, Okagaki R, et al. An algorithm for the management of coagulopathy from postpartum hemorrhage, using fibrinogen concentrate as first-line therapy. *Int J Obstet Anesth* 2017;32:11-6.
78. Woman Trial Collaborators. Effect of early tranexamic acid administration on mortality, hysterectomy, and other morbidities in women with post-partum haemorrhage (WOMAN): an international, randomised, double-blind, placebo-controlled trial. *Lancet* 2017;389:2105-16.
79. World Health Organization. Updated WHO recommendation on tranexamic acid for the treatment of postpartum haemorrhage. Geneva: WHO; 2017.
80. Marshall AL, Durani U, Bartley A, et al. The impact of postpartum hemorrhage on hospital length of stay and inpatient mortality: a National Inpatient Sample-based analysis. *Am J Obstet Gynecol* 2017;217:344 e1-6.
81. Dobson GP, Doma K, Letson HL. Clinical relevance of a p-value: Does TXA save lives after trauma or postpartum hemorrhage? *J Trauma Acute Care Surg* 2018;84:532-6.
82. Lier H, Maegele M, Shander A. Tranexamic acid for acute hemorrhage: a narrative review of landmark studies and a critical reappraisal of its use over the last decade. *Anesth Analg* 2019;129:1574-84.
83. Sentilhes L, Madar H, Mattuizzi A, et al. Tranexamic acid for childbirth: why, when, and for whom. *Expert Rev Hematol* 2019;12:753-61.
84. Sentilhes L, Winer N, Azria E, et al. Tranexamic acid for the prevention of blood loss after vaginal delivery. *N Engl J Med* 2018;379:731-42.
85. Ker K, Shakur H, Roberts I. Does tranexamic acid prevent postpartum haemorrhage? A systematic review of randomised controlled trials. *BJOG* 2016;123:1745-52.
86. Li C, Gong Y, Dong L, et al. Is prophylactic tranexamic acid administration effective and safe for postpartum hemorrhage prevention? A systematic review and meta-analysis. *Medicine (Baltimore)* 2017;96:e5653.
87. Xia Y, Griffiths BB, Xue Q. Tranexamic acid for postpartum hemorrhage prevention in vaginal delivery: a meta-analysis. *Medicine (Baltimore)* 2020;99:e18792. ■